THERON GUO

PhD Candidate in Computational Science

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EXECUTIVE SUMMARY

A Computational Scientist at the Eindhoven University of Technology developing reduced order models for non-linear problems in computational solid mechanics. Integrates data-driven techniques with physics-based modeling to combine the best out of both, proven by an established track record including 2 peer reviewed journal publications. Future research directions include the design of materials with optimal properties for specific target applications.

EXPERIENCES

PhD Candidate

Eindhoven University of Technology

Feb 2020 - Present | Eindhoven, The Netherlands

- Development of reduced order models for highly non-linear multi-scale problems
- Implementation of non-linear finite element solver for microscopic simulations
- Contact person for high performance computing and responsible for web page of research group
- Keywords: principal component analysis, Gaussian process regression, non-linear finite element analysis, data science, dimensionality reduction, machine learning, artificial intelligence

Graduate Research Assistant

RWTH Aachen University

Jan 2019 - Oct 2019 | Aachen, Germany

- Implementation of cohesive zone element into FEAP for fracture and contact mechanics
- Simulation of thermo-mechanical model and validation with experimental results
- Keywords: FEAP, cohesive zone elements, FORTRAN, fracture mechanics, contact mechanics, augmented Lagrangian method, multi-physics, ceramic composites

Visiting International Research Student

University of British Columbia

Jun 2017 - Aug 2017 | Vancouver, Canada

- Simulation of problems of fracture mechanics with non-local peridynamical formulation
- Comparison of peridynamical theory with continuum formulation
- Keywords: LAMMPS, C++, fracture mechanics, additive manufacturing, composite materials, peridynamics

Intern in Interdisciplinary Simulation and Optimization

Robert Bosch GmbH

Aug 2016 - Dec 2016 | Bühl, Germany

- Investigation of the reliability of engineering components
- Development of interface between MATLAB and Python
- Application of genetic algorithms for component optimization
- Keywords: non-intrusive spectral projection, Monte-Carlo simulation, stochastic analysis, Ansys OptiSLang, multi-objective optimization

EDUCATION

M.Sc. in Computational Engineering Science (cum laude)

RWTH Aachen University

Oct 2017 – Sep 2019 | Aachen, Germany

B.Sc. in Computational Engineering Science (cum laude)

RWTH Aachen University

Oct 2013 – Sep 2017 | Aachen, Germany

Partner University Exchange Program

University of Tokyo

Apr 2018 – Aug 2018 | Tokyo, Japan

Erasmus+ Exchange Year

Imperial College London

Oct 2015 – Jun 2016 | London, UK

TECHNICAL SKILLS

- Python
- C++
- MATLAB
- Julia
- Linux

SOFTWARE/PACKAGES

- PyTorch
- ABAQUS
- MOOSE
- Gmsh

LANGUAGES

- German
- English
- Chinese
- Dutch
- Japanese

TEACHING EXPERIENCES

Instructor in Calculus and Linear Algebra

Nov 2020 - Present | Eindhoven University of Technology

WORKSHOPS

Data Assimilation Spring School

Mar 2022 | University of Potsdam

Big Data Summer School

Jul 2016 | Tsinghua University

PUBLICATIONS

Journal Articles

- Guo, T., Rokoš, O., & Veroy, K. (n.d.). Hyper-reduction of geometrically parameterized non-linear microstructures. *In Preparation*.
- Guo, T., Silva, F. A., Rokoš, O., & Veroy, K. (2022). Learning constitutive models from microstructural simulations via a non-intrusive reduced basis method: Extension to geometrical parameterizations. *Computer Methods in Applied Mechanics and Engineering*, *401*, 115636.
- Guo, T., Rokoš, O., & Veroy, K. (2021). Learning constitutive models from microstructural simulations via a non-intrusive reduced basis method. *Computer Methods in Applied Mechanics and Engineering*, *384*, 113924.

SELECTED CONFERENCE TALKS

1. "Hyper-reduction of geometrically parameterized nonlinear microstructures", Model Reduction and Surrogate Modeling (MORE), September 19 – September 23, 2022
2. "A PDE-Based Transformation Method for Model Order Reduction of Nonlinear Geometrically Parameterized Microstructures", 15th World Congress on Computational Mechanics, July 31 – August 5, 2022
3. "Learning Constitutive Models with a Non-intrusive Reduced Basis Method ", 16th U.S. National Congress on Computational Mechanics, July 25 – July 29, 2021

AWARDS

ECCOMAS Scholarship
Scholarship for attending ECCOMAS Congress 2022

SFB1294 Fellowship
Scholarship for spring school on Data Assimilation

DAAD PROMOS
Scholarship for exchange semester at the University of Tokyo

RWTH Aachen University Dean's List
Among top 5% of students

DAAD RISE
Scholarship for research internship at the University of British Columbia

Big Data Summer School Fellowship
Scholarship for attending summer school in Tsinghua University

ERASMUS+ Scholarship
Scholarship for exchange year at Imperial College London

REFERENCES

Prof. Dr. Karen Veroy
@ Eindhoven University of Technology
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Dr. Ondřej Rokoš
@ Eindhoven University of Technology
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